

key is extremely difficult, even if the corresponding public key is known to an attacker.

Steganography means 'covered writing' in Greek; it is the science of writing hidden messages so that no one besides the sender and intended recipient suspects there is a message—security through obscurity. Digitally, information could be hidden in computer files (a document file, image file, program, etc) or transport layers. For example, a sender might take an image file and adjust the colour of every fiftieth pixel to indicate a letter of the alphabet. Someone not specifically looking for this kind of modification is unlikely to notice it.

Later in the article, we will see how steganography can be made far more useful by combining it with cryptography—thus, if the hidden message is discovered by an eavesdropper, they still have the burden of trying to decrypt the message.

The platform

Python, itself a flexible and productive programming language, has the popular libraries EzPyCrypto and Stepic available. EzPyCrypto is a very simple API for military-grade cryptography in Python. It encrypts and decrypts arbitrary-sized pieces of data like strings or files with asymmetric key cryptography.

Stepic provides a Python module and also a command-line interface to hide arbitrary data within images. It slightly modifies the colours of pixels in the image to store the data. These modifications are imperceptible to humans but can be detected by programs.

Setting up the libraries

Practically all Linux distributions will have Python installed, or available as installable packages. If you need to install Python itself, visit the *References* section for a URL.

Download the *ezPyCrypto-0.1.1.tar* package (see *References* for the URL) and run these commands as the root user:

```
# tar xvf ezPyCrypto-0.1.1.tar
# cd ezPyCrypto-0.1.1
# python setup.py install
```

To set up Stepic, you need to install related packages: *Imaging-1.1.6.tar.gz* (the Python Imaging Library); *libpng10-1.0.42-1.fc11.i386.rpm*; *xv-3.10a-13.i386.rpm*. To use the

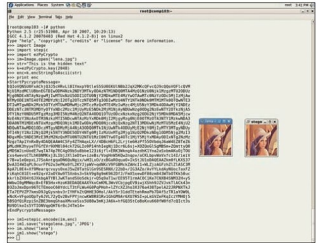


Figure 1: Code, and 'before-and-after' images

show() method of the PIL Image class, we need to install the *xv* utility, which in turn needs the *libpng* library package.

First set up PIL—download the *Imaging-1.1.6.tar.gz* archive, and (as the root) run the following:

```
# gunzip Imaging-1.1.6.tar.gz
# tar xvf Imaging-1.1.6.tar
# cd Imaging-1.1.6
# python setup.py install
```

Next, download Stepic (URL is in *References*) and, as the root, run the following code:

```
# tar xvf stepic-0.3.tar.gz
# cd stepic-0.3
# python setup.py install
```

Steganography with Stepic

Stepic's *Steganographer* class is now deprecated in favour of directly-callable functions in the module. The Stepic page in the *References* section links to API documentation for the module.

1. Start the Python interactive shell:

```
$ python
```

2. Import PIL's *Image*, and the *stepic* module:

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